

1 (a) Describe production of ATP by oxidative phosphorylation. [8]

- 1 - occurs in the inner membrane/cristae;
- 2 - reduced NAD/FAD (from glycolysis/krebs cycle);
- 3 - passed to electron transport chain;
- 4 - hydrogen released from reduced NAD/FAD/reduced NAD/FAD oxidised;
- 5 - hydrogen split into electrons and protons;
- 6 - electrons pass along electron carriers/cytochromes;
- 7 - from one carrier at a higher energy level to another at low energy level;
- 8 - energy is released; ⁹protons pumped into inter-membranal space; ¹⁰proton gradient generated;
- 11 - ref ATP synthase/stalked particles; ¹¹protons flow back into matrix;
- 12 - their electrical potential energy is used to produce ATP from ADP and inorganic phosphate;
- 13 - ref to oxygen as final electron/acceptor ¹³Hydrogen acceptor [12 max 8]

(b) Relate, the following properties of water to their importance to living organisms:

- (i) transparent [2] 3 max 2
- (ii) high heat of vapourisation [2] 3 max 2
- (iii) ice is less dense [2] 4 max 2

Transparent → ¹⁴allow light to penetrate; for photosynthetic aquatic organisms; ¹⁵allows vision (of food/predator)

High heat of vaporisation → ¹⁶large amount of heat can be lost; ¹⁷with minimal loss of water from the body;

Ice is less dense → ¹⁸water evaporates resulting in cooling;

¹⁹ice floats and insulate water below;

²⁰increasing survival chances of aquatic organisms;

²¹circulation of nutrients; ²²colonization of water bodies to greater depth [8 max 6]

(c) Distinguish between starch and glycogen. [6]

Starch	Glycogen
²⁴ consists of 2 forms - (amylose and amylopectin)	- consists of one form - similar to amylopectin;
²⁵ - glucose - helical coiling (in amylose)	- glucose; - no helical coiling
- 1;4 and 1;6 glycosidic bonds	- 1;4 and 1;6 glycosidic bonds;
²⁶ - energy store in plants	- energy store in animals/fungi;
²⁷ - less branched	- more branched;
²⁸ - do not hydrolyse easily	- hydrolyses easily;
²⁹ - starch stored as grain	- stored as granules;

[6 max 6]

2

(a) Describe the behaviour of chromosomes during prophase of meiosis 1. [8]

- 1 - shorten and thicken; *condense*
- 2 - become visible;
- 3 - homologous chromosomes pair/synapsis;
- 4 - form a bivalent;
- 5 - chiasmata formation;
- 6 - homologous chromosomes partially repel each other;
- 7 - forming various shapes;
- 8 - crossing over occurs;
- 9 - between non sister chromatids;
- 10 - swapping of genes/portions;

[10 max 8]

(b) Explain, using one example, the effect of mutations on the phenotype. [6]

- mostly point/gene mutation;
- new alleles;
- e.g. named;
- change codons;
- different proteins produced;
- ref to metabolism;
- expresses;

(c) Distinguish between DNA replication and DNA transcription. [6]

- 26 - enzyme DNA polymerase involved in replication of DNA, while RNA polymerase involved in transcription;
- 27 - DNA nucleotides used in DNA replication while RNA nucleotides are used in transcription; *ref to complementary base pairing (A-U, A-T)*
- 28 - both strands act as template in replication while one strand acts as template in transcription;
- 29 - double stranded molecule produced during replication while single stranded molecules produced during transcription;
- 30 - DNA molecule produced during replication while RNA molecule produced in transcription;
- 31 - molecules produced are identical in replication while a different molecule is produced in transcription;
- 32 - whole molecule is copied in replication while cistron/DNA portion is copied;
- 33 - *DNA ligase in repln, absent in transcription*
- 34 - *replication continuous & discontinuous while transcription is always continuous*

[7 max 6]

3

(a) Describe the fixation of carbon dioxide in C4 plants. (10 max 6)

- 1 - accepted by PEP;
- 2 - ref PEP carboxylase;
- 3 - forming oxaloacetate;
- 4 - oxaloacetate converted to malate;
- 5 - malate shunted into bundle sheath cell;
- 6 - through plasmodesmata;
- 7 - carbon dioxide released by malate;

- 8 - forming pyruvate;
- 9 - increasing carbon dioxide concentration;
- 10 - pyruvate recycled to mesophyll cells;
- 11 - carbon dioxide accepted by RUBP;
- 12 - RUBP carboxylase;

[12 max [8]

(b) Relate the structure of a mitochondrion to its function.

[6]

- Matrix is
- 13 - site of link ^{rxn} and krebs cycle;
 - 14 - contains enzymes for link ^{rxn} and krebs cycle;
 - 15 - contains coenzymes/NAD/FAD molecules;

Mitochondrial (circular) DNA and ribosome

- 16 - produce mitochondrial protein;
- 17 - for replication of mitochondrion;

External/outer membrane

- 18 - allow passage of molecules into and out of mitochondrion;

Inner membrane

- 19 - folds into ^{forms} cristae;
- 20 - large surface area for electron carrier;
- 21 - contain ATP synthase;
- 22 - for ATP synthesis;
- 23 - impermeable to protons / H^+ allowing buildup of proton gradient;
- 24 - intermembrane space to allow accumulation of protons;

[9 max 6]

(c) Discuss the need for genetic counselling in genetic screening.

[6]

- 25 - provide information; about risk of screening / severity of condn
- 26 - making an accurate diagnosis of genetic condition;
- 27 - interpreting results of genetic screening;
- 28 - investigating family history for previous genetic condition;
- 29 - calculating risk of having affected children; helps ptntl parents to decide whether to have children or not;
- 30 - explain cause of genetic condition;
- 31 - quality of life of affected child discussed;
- 32 - providing options to deal with genetic condition;
- 33 - e.g. gene therapy/abortion/IVF/adoption etc; treatment;
- 34 - sharing responsibility for causing the genetic condition; responsibility if a child is a vet
- 35 - to share on availability of support grps
- 36 - ref to life expectancy

[9 max 6]

4

(a) Describe the effects of deforestation on ecosystems.

[8]

- 1 - soil prone to erosion;
- 2 - reduced soil fertility;
- 3 - siltation of rivers and dams;
- 4 - loss of wildlife habitats;
- 5 - carbon dioxide sinks reduced;
- 6 - increased atmospheric CO_2 ;
- 7 - leading to global warming;
- 8 - loss of biodiversity;
- 9 - eutrophication; loss of food sources

10 - land degradation;

(10 max 8)

(b) Distinguish, ~~using a table~~, between eukaryotic and prokaryotic cells. [6]

- 11 - cell size of prokaryotes is 0,5 – 10µm whilst in eukaryotes is 10 – 100µm;
- 12 - DNA in prokaryotes is circular while in eukaryotes DNA is linear;
- 13 - DNA is ~~naked~~ in prokaryotes whilst is contained in nucleus in eukaryotes;
- 14 - prokaryotes have 70s ribosomes whilst eukaryotes have 80s ribosomes;
- 15 - prokaryotes have no ϵr but eukaryotes have ϵr ;
- 16 - few/no membrane bound organelles in prokaryotes whilst many/membrane bound organelles in eukaryotes;
- 17 - in prokaryotes flagella are simple/without microtubule/extracellular/20nm in diameter whilst in eukaryotes flagella are complex/with 9 + 2 arrangement of microtubules/intracellular/200nm in diameter;
- 18 - mesosome present for respiration in prokaryotes whilst mitochondria present in eukaryotes;
- 19 - prokaryotes have photosynthetic membranes for photosynthesis whilst eukaryotes have chloroplasts;
- 20 - cell wall made of murein (prokaryotes) - cellulose/chitin if present; [9 max 6]
- 21 - ~~ref to histone proteins~~ - non can find; 11

(c) Explain the biological advantages of using human insulin from gene technology. [6]

- 22 - large quantities can be prod to meet demand
- 23 - purer than animal insulin/no contamination;
- 24 - identical to human insulin;
- 25 - no immune rejection / no allergic effects / damaging side effects
- 26 - more rapid response;
- 27 - a shorter duration of response;
- 28 - effective to people who are tolerant to animal/derived insulin; (10 max 6)
- 29 - acceptable to people against use of animal insulin / fewer ethical issues
- 30 - reduces slaughtering of cows/pigs;
- 31 - cheaper / easier to produce / purify

5 (a) Describe how an action potential is generated. [8]

- 1 - (membrane) at resting potential;
- 2 - of -65 to -70mv;
- 3 - inside is negative relative to outside of axon/high sodium ions outside and high potassium ions inside;
- 4 - stimulation of cell;
- 5 - opening of sodium ion channels;
- 6 - sodium ions diffuse into cell;
- 7 - membrane depolarised;
- 8 - to threshold value of -50mv/-55mv;
- 9 - positive feedback;
- 10 - causes opening of more sodium ions channels;
- 11 - influx of sodium ions into cell;
- 12 - causing further depolarisation;
- 13 - potential difference reaches + 40mv;
- 14 - sodium ions channels close;

[14 max 8]

(b) Compare, using a table, the structure of an artery and a vein.

[6]

- 15 - both have tunica externa;
- 16 - both have tunica media;
- 17 - both have tunica intima/endothelium;
- 18 - both have walls impermeable to blood;
- 19 - arteries have no valves while veins have valves;
- 20 - artery have a small lumen while veins have large lumens;
- 21 - thick tunica media on artery and thin tunica media on vein; *more elastic fibres / less*
- 25 - *thicker tunica externa on vein / thinner on artery* [7 max 6]
- 26 - *folded tunica intima in artery / smooth in vein*

(c) Explain any three long-term benefits of exercise to cardiovascular system.

[6]

- 22 - *increase efficiency of cardiovascular system; vascularization*
- 23 - *increase in supply of blood to muscles;*
- 24 - *increase in cardiac output;*
- 25 - *increase in stroke volume;*
- 26 - *more blood pumped per beat;*
- 27 - *fall in resting heart rate;*
- 28 - *heart muscle increase; stronger / size of heart increases*
- 29 - *more efficient at pumping blood;*
- 30 - *increase in size of the heart chamber;*
- 31 - *decrease in resting systolic and diastolic pressures;*
- 32 - *risk of BP reduced; hypertension / CHDs*
- 33 - *powerful heart beat;*
- 34 - *increase in number of mitochondria on walls of the heart;*
- 37 - *any correct explanation; size* [10 max 6]

6

(a) Describe the development of a plant embryo sac into an ovule.

[8]

- 1 - *diploid mother cell undergoes meiosis (to produce 4 haploid cells)*
- 2 - *3 haploid cells disintegrated*
- 3 - embryo sac grows;
- 4 - its nucleus undergo repeated mitosis;
- 5 - eight nuclei are produced/four at each end of the embryo sac;
- 6 - two (polar) nuclei migrate to the centre of embryo sac;
- 7 - fuse to become a (single) diploid nucleus;
- 8 - remaining 6 nuclei become separated by the cell walls;
- 9 - 5 nuclei (3 antipodal and 2 synergids) disintegrate; *(10 max 8)*
- 10 - Female gamete remains;

(b) Explain any three risks of abortion.

[6]

- 11 - occasional infections occur;
- 12 - leading to blockage of oviducts;
- 13 - causing infertility;
- 14 - ectopic pregnancy risks increase;
- 15 - damage to the cervix;
- 16 - reduced ability to carry another baby;
- 17 - placenta may be retained;
- 18 - causing excessive bleeding;
- 19 - may lead to death;

9 max 6

- (c) Explain how the structure of the placenta is adapted to exchange of materials between maternal and foetal blood. [6]

- 20 - chorionic villi;
- 21 - increase surface area for exchange of material by diffusion;
- 22 - numerous mitochondria;
- 23 - for ATP production for active transport/pinocytosis;
- 24 - capillary network in foetal part;
- 25 - increase surface area for exchange;
- 26 - short distance between foetal and maternal blood systems;
- 27 - speed up exchange of materials;

8 max 6

- 7 (a) Describe the importance of Kingdom Plantae. [8]

- 1 - producer/carbon sinks; *reduce atmospheric CO₂ / reduce global warming*
- 2 - source of medicine;
- 3 - source of timber;
- 4 - beautifying the environment;
- 5 - balance of nature in an ecosystem; *producers;*
- 6 - source of food; *symbiotic*
- 7 - form parasitic relationships with other organisms e.g. fungi;
- 8 - habitats for other organisms e.g. birds;
- 9 - pioneer colonisers of new habitats;
- 10 - provide ground cover;
- 11 - *produce oxygen; beliefs;*
- 12 - *ref to cultural beliefs;*

[10 max 8]

- (b) Suggest any six reasons for maintaining biodiversity. [6]

- 13 - ecotourism;
- 14 - medicinal source;
- 15 - source material/timber;
- 16 - for future use;
- 17 - food;
- 18 - maintain balance in ecosystem;
- 19 - maintain a gene pool/source of genes;
- 20 - prevent extinction;
- 21 - for beauty;
- 22 - organism has right to survive;
- 23 - source of foreign currency;
- 24 - *humans have a responsibility to maintain biodiversity / stewardship*

(12 max 6)

- (c) Explain the importance of elephants in a natural ecosystem. [6]

- 25 - open forest canopies;
- 26 - light penetration;
- 27 - leading to prompt growth of grasses;
- 28 - providing food for grazers;
- 29 - opens tracks for other animals;
- 30 - pull down leaves and branches;
- 31 - so other animals can reach/for browsers;
- 32 - dig water holes in dry river beds;
- 33 - other animals can benefit;

13 max 6

- 34 - provide dung which is food for beetles;
 35 - providing nutrition for saprophytes;
 36 - dung improves soil fertility;
 37 - seed dispersal

[12 max 6]

2b)

- 11 - gene mutation;
 12 - named e.g. ; PKU / S.C.A /
 13 - change in gene / DNA / base change / nucleotide change in DNA; [8 max 6]
 14 - detail of change e.g. substitution / addition / deletion / frameshift;
 15 - different mRNA prodn / different codons;
 16 - different amino acid coded for;
 17 - different polypeptide / protein / non-functional prtn prodn;
 18 - named effect of mutation; (Trisomy 21) / Turner's Syndrome / Klinefelter syndrome;
 19 - chromosome mutation; / inversion / translocation / duplications / deletions;
 20 - named e.g. Down's Syndrome (Trisomy 21) / Turner's Syndrome / Klinefelter syndrome;
 21 - structural changes in chromosomes / inversion / translocation / duplications / deletions;
 22 - change in chromosome No / ~~aneuploidy~~; [6 max 6]
 23 - change in sets of chrms / polyploidy
 24 - named effects;
 25 - ref to non-disjunction / aneuploidy